University of Galway

Electrical & Electronic Engineering Department

Project Report Template

Version 2.1

March 2026

Report Format:

The final Project Report is an important document that describes the problem statement, the motivation, the background, what work has been accomplished and how that work was carried out during your project work, results and analysis.

For 2026 the report format is modified to be shorter and more focussed. We include the details of what a full project report would normally include but you do not need to cover all sections in your report. Those shaded in orange are not required. Instead you should provide a single Introduction/Background section *[max 5 Pages]* ; the bulk of the report should focus on technical sections that details any design elements, the implementation of your project and how you tested/validated it *[max 20 Pages]*. Any experimental details should be recorded and results presented graphically or in tabular form as appropriate. Finally you should have a concluding chapter/section that summarizes your achievements, reflects on the project experience and what you've learned from this and possibly include some recommendations for future work *[max 5 Pages]*.

The following should roughly comprise the structure of your report. Note that these are *guidelines*, not *rules*. You have to use your intelligence in working out the details of your specific writing. Not all elements listed here require a separate section – use your own judgement, but try to address any that are important in the context of your project.

1. **Cover page:** You should have a proper cover page that mentions the project title, name of the course, semester, team members, supervisor name, and date of submission.
    - The title should reflect what you have done and should bring out any eye-catching factor of your work, for good impact.

2. **Abstract:** The abstract should be short, generally within about 2 paragraphs (250 words or so total). The abstract should contain the essence of the report, based on which the reader decides whether to go ahead with reading the report or not. It can contain the following in varying amounts of detail as is appropriate: main motivation, main design point, essential difference from previous work, methodology, and some eye-catching results if any.
3. **Table of Contents:** Insert a Table of Contents which lists all the sections of your report (upto 3 levels – Heading1, Heading2 and Heading3) against page numbers. Go to Reference and select Table of Contents to insert one, and format all section/sub-sections titles appropriately as Heading 1, 2 or 3. Don't forget to refresh the Table of Contents every time you make changes to your report.
4. **List of Tables and Figures:** Include a complete list of all tables and figures in your report.
5. **Introduction:** *[recommended max 5 Pages for 5 + 6]* This section is a shorter version of the rest of the report. After abstract, introduction and conclusions are the two mainly read parts of a report. In this section you should answer the following questions *(note that you don't need to explicitly follow the list below, or include all items in the list; use your own judgment on which is relevant to your project)*:

- What exactly is the problem you are trying to solve? *This is the problem statement.*
- Why is the problem important to solve? This is the *motivation*. In some cases, it may be implicit in the problem statement itself.
  - Is the problem still unsolved? *This is a statement of past/related work.*
  - Why is the problem difficult to solve? *This is a statement of challenges*…
  - How have you solved the problem? Here you state the essence of your *approach*. This can be expanded upon later, but it must be stated explicitly here.
- What are the conditions under which your solution is applicable? *This is a statement of assumptions*.
- What are the main results? Present the main *summary of the results* here.
- What is the summary of your contributions?
- How is the rest of the report organized? Here you include a paragraph on the *flow of ideas* in the rest of the report.

6. **Background:** In this section include sufficient background about the basic aspects or the theoretical underpinnings of your project which the general reader must understand before knowing the details of your work. *[For 2026 combine with your Introduction section]*
7. **Past/related work:** In this section explain what is novel about what you have done in this project. Here, you must try to think of *dimensions of comparison* of your work with other work. For instance, you may compare in terms of functionality, in terms of performance, and/or in terms of approach.
8. **Project Management:** In this section, you must describe your project management strategy – how you have managed this project given the resources and time (project duration) in order to successfully complete this project. List the break-down of your project into a set of smaller tasks, role of team members and describe how they contributed to the success of the project. Show your project plan using Gantt Charts and/or Pert Charts.

   Also, explain any changes in the project plan from the original one in your proposal, if any. It is normal to make changes to your original project plan as the project progresses towards its completion. In a nutshell, provide a summary of the main project plan developed in the early stages of the project and reflect on how this evolved over time and what you have learned about managing your project work during the year.

9. **Technical Sections:** *[recommended max 10 Pages]* This is the most important section as it describes your design in details. You may have different sections which delve into different aspects of the design. The organization of the report here is problem specific. The organization of the report here is problem specific. You may also have a separate section for statement of design methodology, or experimental methodology.

   You should mention the following main points:

   - **Design details:** Describe your design in as much detail as possible supported by pictures, graphs and drawings.
   - **Calculations**: Include all necessary calculations as well as design/circuit diagrams. Make sure your figures are clear and well labeled.
   - **Implementation details:** Describe how you implemented the design elements of your project; this should include aspects of any hardware, software or algorithm functionality as appropriate. It is
   - **Communication & Style Tips:**
     - **Use of figures:** The saying "a picture is worth a thousand words" is appropriate here. Spend time thinking about pictures. Wherever necessary, explain all aspects of a figure and do not leave the reader wondering as to what the connection between the figure and the text is (ideally include a detailed explanation of the figure in the main text!).
     - **Terminology***:* Define each term/symbol before you use it, or right after its first use. Stick to a common terminology throughout the report.

10. **Project Results and Analysis:** *[recommended max 10 Pages]* In this section, discuss the project results in detail supported by facts and figures.

    - State the principal results and discuss them.
    - If you have proposed a new idea, algorithm or design, discuss how it compares with existing ones.
    - Tabulate your data and produce necessary plots. Analyze the data and/or plots and make comments.
    - Depending on your project you might describe system testing & validation aspects in this section.

11. **Challenges Faced:** Describe all challenges you faced in the project and how they impacted the project schedule, scope or expected results. Also, write about how you solved these challenges or found a way around them. (Note that this aspect of the report might not require a separate section, and might fit better if included in the Project Planning, Technical Details or Project Results & Analysis sections).

    - Problems with team members not cooperating/meeting.
    - Problems or delays in procuring required parts/components/tools.
    - Problems with equipment or components not working or malfunctioning.

12. **Final Budget:** In this section, you list all the expenses incurred in completing your project for things such as hardware, software, and related costs, etc. If you exceeded the original estimated budget, give details about and justification for the extra expenses.

13. **Project Deliverables:** In this section, you list out the **actual** project deliverables (results or outcomes) such as a product, detailed design documents, a final report, publications, etc.

The success of the project is gauged from what you actually delivered compared with what you set out to achieve at the beginning of the project. Provide justification if you fell short.

14. **Recommendations and Future Work:** Here you state aspects of the project you have not considered due to lack of time and/or resources. Also, provide suggestions/recommendations for further extensions or improvements.

15. **Conclusions**: *[recommended max 5 Pages]* This is one of the sections commonly looked at by the readers. State briefly the main take-away points from your work. Re-state the main objectives of your project and to what degree they were achieved. Write about what went wrong and your project was impacted. Also, write about the significance of your work and the results you obtained.

16. **References/Bibliography:** All sources of information for your report should be cited in the writing.

    At the end of the report/paper, include a section called References and list each. The titles of books and articles are also to be included with the page numbers. This is intended to provide convenient means for the reader to locate the cited material.

17. **Appendix**: In this section *[Consult with your supervisor]*, you include additional information related to your project which the reader can refer to if needed, such as:

    - data sheets of components you have used
    - source code of the project *[For 2026 code is provided in the project Gihub]*
    - details or descriptions of datasets used (e.g. to train AI models)
    - proofs of theorems, full-page graphs and plots
    - additional PSpice and/or Matlab simulation results, etc.
    - printout of LabVIEW VIs
    - hardware schematics
    - other technical information that is helpful to understand your project work

Cover Page: Your final report must have a cover page with the following information:

- Name of department and university
- Project title
- Names and IDs of group members
- Date of submission
- Supervisor name

Declaration on use of AI: New for 2026; details to follow shortly.

See next page for a sample cover page.

Department of Electronic & Electrical Engineering

National University of Ireland Galway

PROJECT REPORT (Year-5)

Design of a Smart Home System

Group members:

Paddy Smith - 201010101

Jane Doe - 201010102

Date of submission: 22/4/2012

Supervisor: Prof. Johnny Rockett